Jeffrey Taylor

HSTP 160 - Beers of the World

The Deep Chemistry of Beer

Beer is described as a commonly simple beverage made from four main ingredients: water, malted barley, hops, and yeast. This simplicity is largely misleading. In reality, beer is a chemically complex system containing hundreds of volatile and nonvolatile compounds that influence its flavor, aroma, and overall quality (Piornos et al., 2023) . These compounds are not static, they are dynamic, formed and transformed through reactions during brewing and continue to change over time. Beer needs to be understood by brewers not simply as a beverage, but as a system in which molecular interactions, processing conditions, and time-dependent reactions directly determine flavor, stability, and overall quality.

At a molecular level, beer is composed of a wide range of substances, including sugars, alcohols, proteins, acids, and a variety of minerals. These components originate primarily from malted barley and are modified throughout the brewing process. The composition of beer changes at each stage due to ingredient additions, environmental conditions, and biochemical reactions (Liu et al., 2022). Proteins are especially important in determining the quality of the beer. Derived from both barley and yeast, they influence visual properties such as foam stability, and clarity, and more physical properties such as flavor. During malting and mashing, proteins are broken down into peptides and amino acids, which are then used by the yeast during fermentation. At the same time, carbohydrates are converted into fermentable sugars, allowing yeast to further produce ethanol and carbon dioxide (CO_2). The brewing process itself drives these chemical changes. Mashing breaks down starches and proteins, the heat from boiling introduces additional reactions and flavor formation, and fermentation produces alcohol and a variety of secondary compounds. Each step alters the chemical composition of the beer, altering flavor, visual and aromatic properties. This pushes the thought that beer is not just a simple drink but a dynamic chemical process.

The flavor of beer is primarily determined by its chemical composition. Although hundreds of compounds are present, only a limited number significantly affect aroma and taste due to differences in concentration, and sensory thresholds (Piornos et al., 2023). This means that even small changes in compound concentration can significantly affect flavor, making consistency in brewing conditions critical for maintaining quality.This relationship between chemical composition and perception became clear through direct tasting experience. For example, beers such as Duvel and Rochefort 10 showed strong banana and honey-like flavors consistent with ester compounds produced during fermentation. Similarly, beers such as Ommegang Hennepin showed pronounced banana and fruit-like characteristics, even when overall flavor intensity was relatively low, demonstrating how certain compounds can dominate perception even at small concentrations. In contrast, some beers displayed off-flavors that align with known chemical causes. For instance, Newcastle Brown Ale presented a “skunky” aroma, which is commonly associated with light-struck reactions involving hop compounds . These observations reinforce the idea that flavor in beer is not random, but directly tied to specific chemical compounds and reactions.

Malt is arguably the most important contributor to beer chemistry. It provides fermentable sugars, proteins, amino acids, and a wide range of chemical precursors that influence both flavor and stability. Studies have identified thousands of compounds in malt and beer, including amino acids, lipids, and phenolic compounds that directly affect flavor profiles (Bettenhausen et al., 2018). The malting process introduces huge chemical changes. During kilning, heat causes amino acids and sugars in the malt to react in what is known as a Maillard reaction, a process that produces browning and creates complex flavor compounds responsible for caramel, roasted, and bready notes. These malt-driven flavor changes were also evident in tasting darker beers. For example, beers such as Delirium Nocturnum and Murphy's Stout showed strong chocolate, coffee, and roasted characteristics, which are consistent with compounds formed through Maillard reactions and roasting processes . In particular, the perception of "burnt caramel" and dark chocolate flavors highlights how increased kilning temperatures lead to deeper, more complex flavor development. These reactions depend on temperature and processing conditions, which is why different malts produce different flavor outcomes. Malt composition also affects fermentation. The availability of free amino nitrogen influences yeast metabolism and the production of flavor-active compounds such as esters and diacetyl. Additionally, variations in barley type and growing conditions can lead to measurable differences in beer chemistry.

Beer is not only defined by its chemical composition but also by its physical behavior. One of the most noticeable physical characteristics is the head of the beer, or the foam. When carbon dioxide is released in the beer, they rise to the surface as bubbles forming what we know as the "head". This process depends on factors such as temperature, carbonation level, and the presence of nucleation sites (Bamforth, 2023). Foam stability is determined by a balance between stabilizing and destabilizing compounds. Proteins and hop-derived substances help stabilize foam by forming protective layers around bubbles. In contrast, ethanol and lipids weaken these layers and contribute to foam collapse. The overall stability of foam depends on the relative concentrations of these competing components. Smaller bubbles tend to collapse as gas moves into larger bubbles, reducing foam stability over time. Foam and carbonation differences were also noticeable during tasting. For example, nitrogen-infused beers such as Murphy's Stout produced a thicker, creamier head compared to standard carbonated beers, creating a texture similar to coffee when poured . This reflects how gas composition and bubble size influence mouthfeel and foam stability. In contrast, some lighter beers showed weak or quickly dissipating foam, indicating a lower presence of stabilizing proteins or higher concentrations of destabilizing compounds.

After brewing is complete, beer continues to undergo chemical changes during storage. These changes are primarily driven by reactions such as oxidation, which alter both the composition and sensory properties of the beer. As a result, beer is considered chemically unstable, as a product (Vanderhaegen et al., 2006). One of the most well-known effects of aging is the development of off-flavors, such as the "cardboard" taste associated with certain oxidation products. At the same time, desirable flavors, including fruity and floral notes, tend to decrease in intensity with aging. The effects of chemical aging were also observed in several beers. Some samples exhibited flavors described as "cardboard-like" or dull, which are commonly associated with oxidation. For example, certain beers were described as having a "skunky" or stale character, reinforcing the impact of environmental exposure such as light and oxygen on beer quality .

These real-world observations align with the chemical understanding that beer is not stable over time, and that improper storage conditions can significantly degrade flavor. Storage conditions play a major role in these changes. Higher temperatures accelerate chemical reactions, while exposure to oxygen increases the rate of oxidation. Because of this, controlling storage conditions is essential for maintaining beer quality over time. Understanding the chemistry of beer is not only important from a scientific perspective, but also has direct applications in brewing and product quality. Breweries rely on knowledge of chemical reactions to control how the flavor develops, ensure consistency, and extend the shelf life of the beer. For example, managing fermentation conditions allows brewers to control ester production, while limiting oxygen reduces the formation of stale flavors during storage. In the growing craft beer industry, small differences in ingredient selection and processing can lead to distinct beer styles that fall outside the norm. This makes chemical knowledge a key factor in innovation and quality control.

Beer is a complex and dynamic chemical system shaped by interactions between ingredients, processing conditions, and time-dependent reactions. Its flavor, aroma, and physical properties are the result of molecular transformations that occur during brewing and continue throughout storage. Both chemical and physical processes contribute to the final product, from fermentation-derived flavor compounds to foam stability driven by protein interactions. Understanding beer at a chemical level demonstrates that even a simple beverage involves advanced principles of chemistry and engineering. This perspective highlights the importance of controlling both ingredients and processes to achieve consistent quality, while also showing how small molecular changes can significantly impact the overall sensory experience.

APA References Sited

Bamforth, C. W. (2023). *The physics and chemistry of beer foam:* A review. European Food Research and Technology, 249, 3–11. https://doi.org/10.1007/s00217-022-04134-4

Bettenhausen, H. M., Barr, L., Broeckling, C. D., Chaparro, J. M., Holbrook, C., Sedin, D., & Heuberger, A. L. (2018). *Influence of malt source on beer chemistry, flavor, and flavor stability.* Food Research International, 113, 487–504. https://doi.org/10.1016/j.foodres.2018.07.024

Liu, S., Kerr, E. D., Pegg, C. L., & Schulz, B. L. (2022). *Proteomics and glycoproteomics of beer and wine.* Proteomics, 22, Article 2100329. https://doi.org/10.1002/pmic.202100329

Piornos, J. A., Koussissi, E., Balagiannis, D. P., Brouwer, E., & Parker, J. K. (2023). *Alcohol-free and low-alcohol beers: Aroma chemistry and sensory characteristics.* Comprehensive Reviews in Food Science and Food Safety, *22*, 233–259.

Vanderhaegen, B., Neven, H., Verachtert, H., & Derdelinckx, G. (2006). *The chemistry of beer aging—A critical review*. *Food Chemistry, 95*(3), 357–381.